

Summary

A 15-puzzle is a good example of adapting the principles of recursion to a real-world problem. Recursion can perform a depth-first search on the tree graph of states that a 15-puzzle produces to find the path to a solution state. However, a purely recursive algorithm won't work, which was why we had to make certain adjustments.

The problem arises because a 15-puzzle has a massive number of possible states and doesn't form a DAG. The edges in this graph are undirected, and the graph contains loops, or cycles. Our solving algorithm needs to ensure that it doesn't make moves that immediately undo the previous move, so that it traverses the graph in one direction. It also needs to have a maximum number of moves the algorithm is willing to make before it begins to backtrack; otherwise, the loops guarantee that the algorithm will eventually recurse too much and cause a stack overflow.

Recursion isn't necessarily the best approach for solving sliding-tile puzzles. All but the easiest puzzles simply have too many combinations for a typical laptop to solve within a reasonable amount of time. However, I like the 15-puzzle as an exercise in recursion because it connects the theoretical ideas of DAGs and DFS into a real-world problem. While 15-puzzles were invented over a century ago, the advent of computers provides a rich tool for exploring techniques to solve these amusing toys.